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Returns to education in the Philippines (2005–2022): Declining benefits, increased inequities

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Education is a crucial determinant of an individual's earnings and contribution to economic growth. In the Philippines, it has a significant impact on income distribution, leading to increased inequality. Previous studies have identified education as the most important factor contributing to wage disparities (Luo and Terada 2009) and a significant factor explaining the urban-rural earnings gap (Chua et al. 2015). Amid historical advancements in education (Hossain and Psacharopoulos 1994), recent years have witnessed expanded access across all levels.

This *Policy Note* examines the returns to education in the Philippines from 2005 to 2022, focusing on salaried graduates. It explores trends in returns across educational levels, particularly among college graduates. The analysis reveals a significant decline in college education returns from 2010 to 2022, which coincides with major policy changes, such as the K to 12 reform in 2012 and the enactment of the free tuition law in 2018.

Salient Points:

- *Returns to education in the Philippines have declined but converged across levels of educational attainment levels from 2005 to 2022.*
- *Returns to college education fell most significantly over that period. However, the decline for college graduates was most pronounced for low-wage workers, reversing previous research findings.*
- *Declining returns may be related to increased formal employment, a shift from agriculture to services, job quality, and reduced education quality. These factors suggest the need for targeted interventions.*

Data and methods

Data from the October rounds of the Labor Force Survey (LFS) from 2005 to 2022, focusing on 2005, 2012, 2019, and 2020–2022, were utilized to examine trends in returns to education. The full sample comprises 621,638 individuals, with a subsample of 185,020 salaried graduates aged 25–34. The choice of 2005 and 2019 is deliberate as these years capture the period before and after significant shifts in educational participation occurred. This allows for a comparison of returns to

education in the context of the reforms implemented during this timeframe, as highlighted by Yee (2023). Additionally, including LFS data from 2012 provides a midpoint reference to assess the pace and magnitude of changes in returns to education between 2005 and 2019. The analysis of LFS data from 2020 to 2022 also sheds light on the immediate and short-term effects of the COVID-19 pandemic on the education-labor market nexus.

Returns to education are estimated using ordinary least squares (OLS) regression, following the standard Mincerian wage equation. The dependent variable is log of hourly wages in 2018 prices. To examine the variances in returns across the earnings distribution, conditional quantile regressions (CQR) are employed, enabling a more nuanced understanding of how the benefits of education differ for low-wage and high-wage workers.[1]

[1] While Yee (2023) uses unconditional quantile regressions, the authors argue that conditional quantile regression is more applicable, as this *Policy Note* focuses on the returns to education for specific quantiles rather than the entire wage distribution.

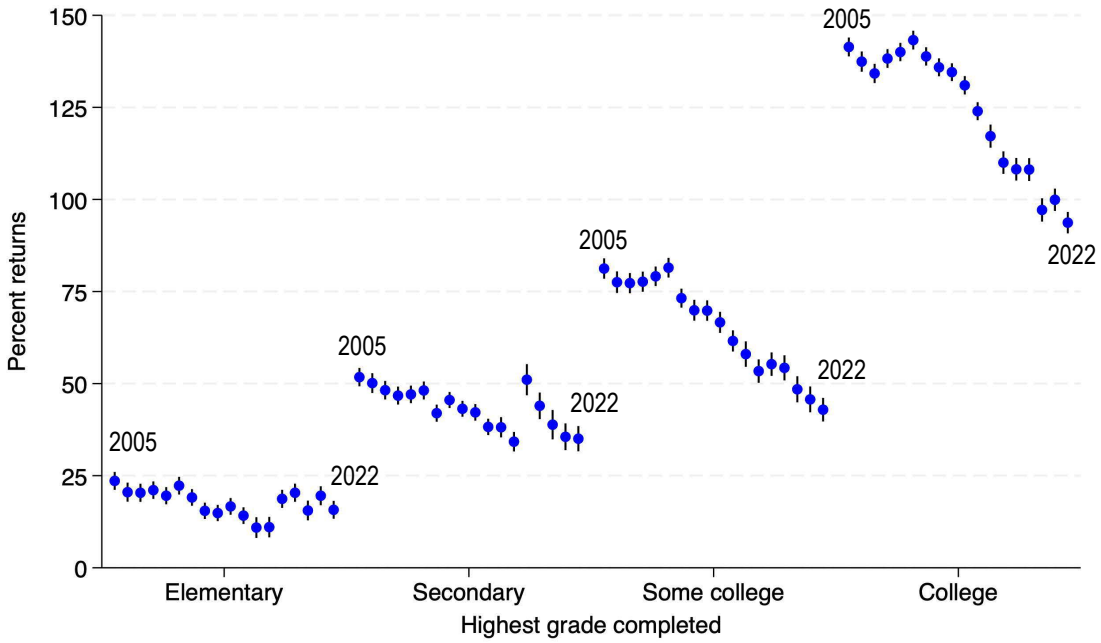
Trends in returns to education in the Philippines

Converging returns across educational levels

Figure 1 presents the OLS returns to the log of hourly wages from 2005 to 2022 for the entire sample. The returns for college graduates declined from 141.4 percent in 2010 to 93.7 percent in 2022, compared to those with no grade completed. Similar declines were observed for partial college and secondary attainment. Returns to elementary education have remained low but stable since 2018. There is a general narrowing of the returns between educational levels, but this is most pronounced between secondary and partial college completion; in 2018, the returns to these two categories were at par.

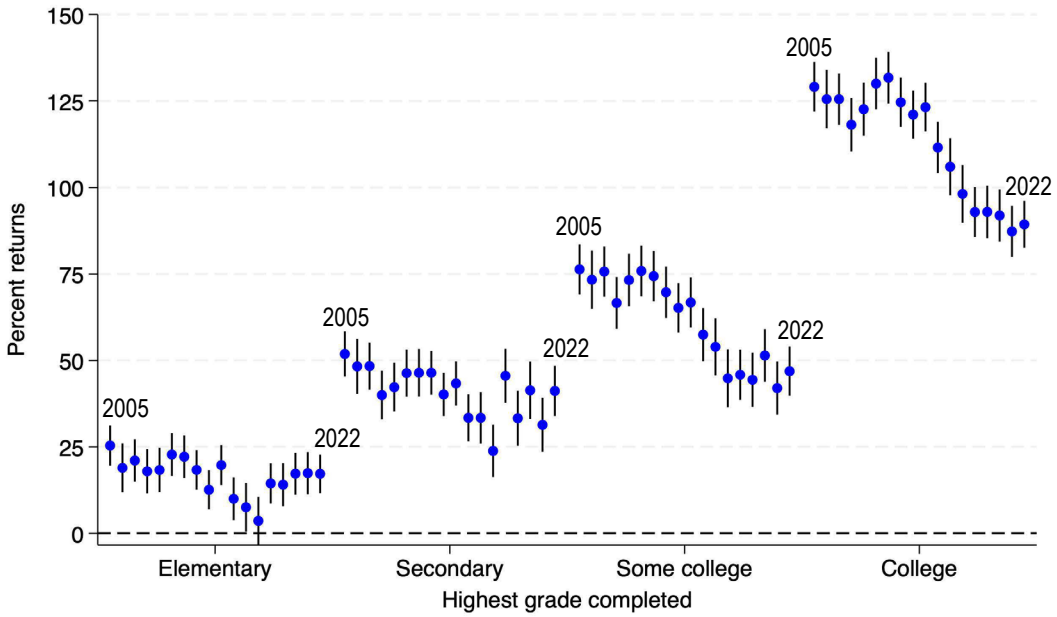
Figure 2 shows the evolution of returns for the young subsample, and similar trends can be observed. The returns to elementary education were zero in 2017, and rose again in 2018 onward. A similar dip in the returns to secondary education was noted in 2017,

Figure 1. Returns to education by highest grade completed, full sample, 2005–2022



Notes: The reference group comprises observations with no grade completed. Regression coefficients are multiplied by 100 percent. Lines represent 95 percent confidence intervals.
Source: Authors' computations from the 2005–2022 October round of the Labor Force Survey (PSA various years)

Figure 2. Returns to education by highest grade completed, young sample, 2005–2022



Notes: The reference group comprises observations with no grade completed. Regression coefficients are multiplied by 100 percent. The lines represent 95 percent confidence intervals. The young sample includes people aged 25–34.
Source: Authors' computations from the 2005–2022 October rounds of the Labor Force Survey (PSA various years)

coinciding with the first batch of senior high school students. There is also a steep decline in returns for college and partial college completion, but the convergence of returns is most pronounced between secondary and partial college completion. Their returns are statistically not different from one another every year since 2018.

Table 1 presents the OLS returns to the log of hourly wages for the full sample from 2005 to 2022. The cumulative drop in the returns to college education during this period was 33.7 percentage points. Meanwhile, Table 2 presents the OLS returns for the young sample, which showed a 30.8 percentage point drop. These findings contrast with the 46 percentage point increase from 2001 to 2010 (Punongbayan 2013).

Both tables show that returns to elementary and secondary education also dropped, with a notable increase between 2017 and 2018 coinciding with the graduation of the first senior high school class. In recent years, secondary education returns have matched the returns to partial college education.

An alternative way of measuring the returns to education is by looking at the years of schooling imputed from respondents' highest grades completed. Using years of schooling, the returns to education from 2005 to 2022 dropped by 2.5 and 3 percentage points for the young and full samples, respectively (Figure 3).

Widening inequities among college graduates

While the overall returns to college education have declined, suggesting a convergence of returns for individuals with varying levels of education, a closer look at the distribution of returns reveals widening inequities among college graduates.

Conditional quantile regressions (Figure 4) show a shift in the returns to college education, with higher returns observed for low-wage workers in 2005, but higher returns for high-wage workers in 2019 and 2022.

Table 1. Returns to various levels of education, OLS estimates, full sample, 2005–2022

	2005	2012	2019	2020	2021	2022
Elementary	0.236***	0.155***	0.204***	0.155***	0.196***	0.157***
	(0.012)	(0.011)	(0.013)	(0.014)	(0.013)	(0.012)
Secondary	0.518***	0.455***	0.440***	0.389***	0.356***	0.350***
	(0.013)	(0.011)	(0.018)	(0.020)	(0.019)	(0.018)
Partial college	0.812***	0.699***	0.543***	0.485***	0.457***	0.429***
	(0.014)	(0.014)	(0.017)	(0.018)	(0.018)	(0.016)
College	1.414***	1.359***	1.081***	0.971***	0.999***	0.937***
	(0.013)	(0.012)	(0.016)	(0.016)	(0.015)	(0.015)
N	30,891	35,988	36,210	32,697	35,272	37,179
R-squared	0.40	0.37	0.34	0.31	0.33	0.31

Notes: *** $p<.01$, ** $p<.05$, * $p<.1$. OLS stands for ordinary least squares. Other controls not shown: sex, experience, squared experience.
Source: Authors' computations from the 2005, 2012, 2019, and 2020–2022 October rounds of the Labor Force Survey (PSA various years)

Table 2. Returns to various levels of education, OLS estimates, young sample, 2005–2022

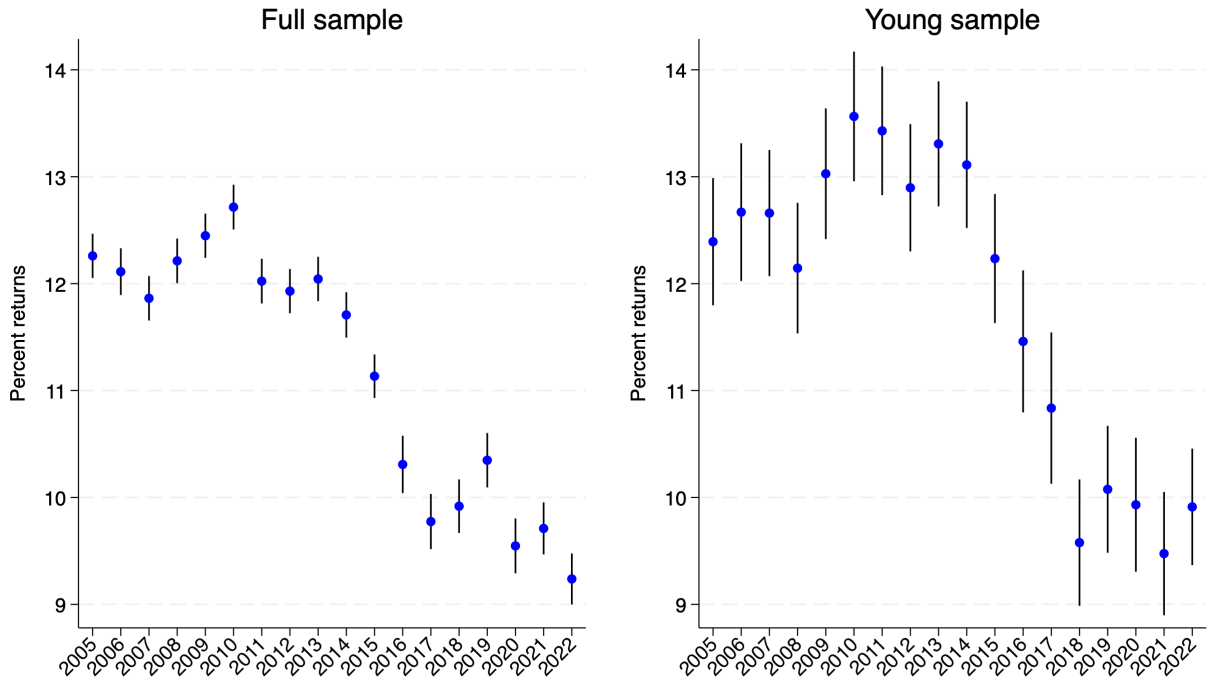
	2005	2012	2019	2020	2021	2022
Elementary	0.254***	0.183***	0.141***	0.172***	0.174***	0.172***
	(0.030)	(0.029)	(0.032)	(0.031)	(0.031)	(0.028)
Secondary	0.518***	0.464***	0.333***	0.413***	0.314***	0.412***
	(0.033)	(0.032)	(0.041)	(0.042)	(0.040)	(0.037)
Partial college	0.763***	0.697***	0.444***	0.514***	0.420***	0.469***
	(0.037)	(0.038)	(0.040)	(0.039)	(0.039)	(0.036)
College	1.291***	1.246***	0.929***	0.919***	0.873***	0.893***
	(0.037)	(0.036)	(0.039)	(0.039)	(0.038)	(0.034)
N	9,355	10,119	11,513	10,169	11,347	11,949
R-squared	0.31	0.33	0.32	0.29	0.30	0.30

Notes: *** $p<.01$, ** $p<.05$, * $p<.1$. OLS stands for ordinary least squares. Other controls not shown: sex, experience, squared experience.
Source: Authors' computations from the 2005, 2012, 2019, and 2020–2022 October rounds of the Labor Force Survey (PSA various years)

Table 3 presents the estimates for the returns to various levels of education by quantile (10th, 50th, 90th). From 2005 to 2022, the 10th quantile experienced a 64 percentage point drop in returns to college education, compared to 18.2 percentage points for the 90th quantile. This reverses previous findings that suggested higher education provided greater benefits for low-wage graduates (Patrinos and Sakellariou 2006). This shift raises concerns about the role of higher education in perpetuating or exacerbating income inequality, as the returns to college education appear to be increasingly skewed in favor of those already at the higher end of the earnings distribution.

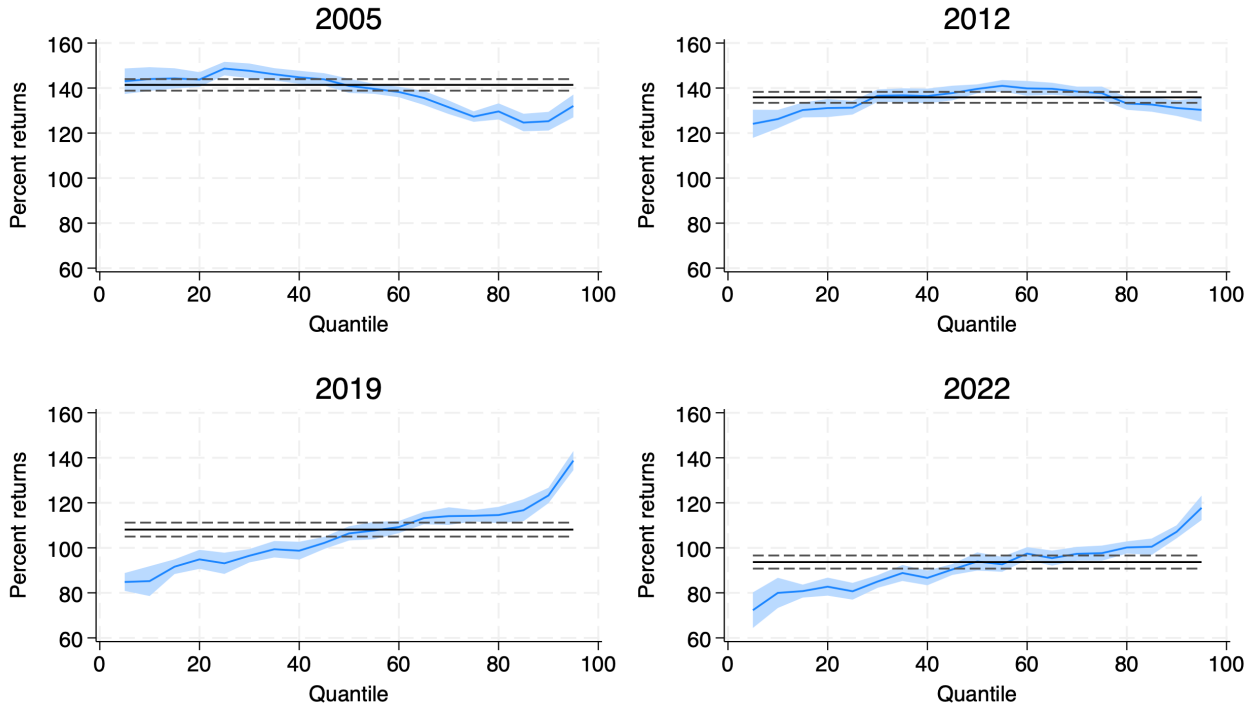
The pivoting of the conditional quantile returns to education is observed as well for the young cohort, even if one uses years of schooling instead of the levels of education (see Figures 5, 6, and 7).

Figure 3. Returns to education using years of schooling, 2005–2022



Note: Regression coefficients are multiplied by 100 percent. Data show percent increase in hourly real wages based on an extra year of schooling. The lines represent 95 percent confidence intervals.
Source: Authors' computations from the 2005–2022 October rounds of the Labor Force Survey (PSA various years)

Figure 4. Returns to college education by quantile, full sample, 2005–2022



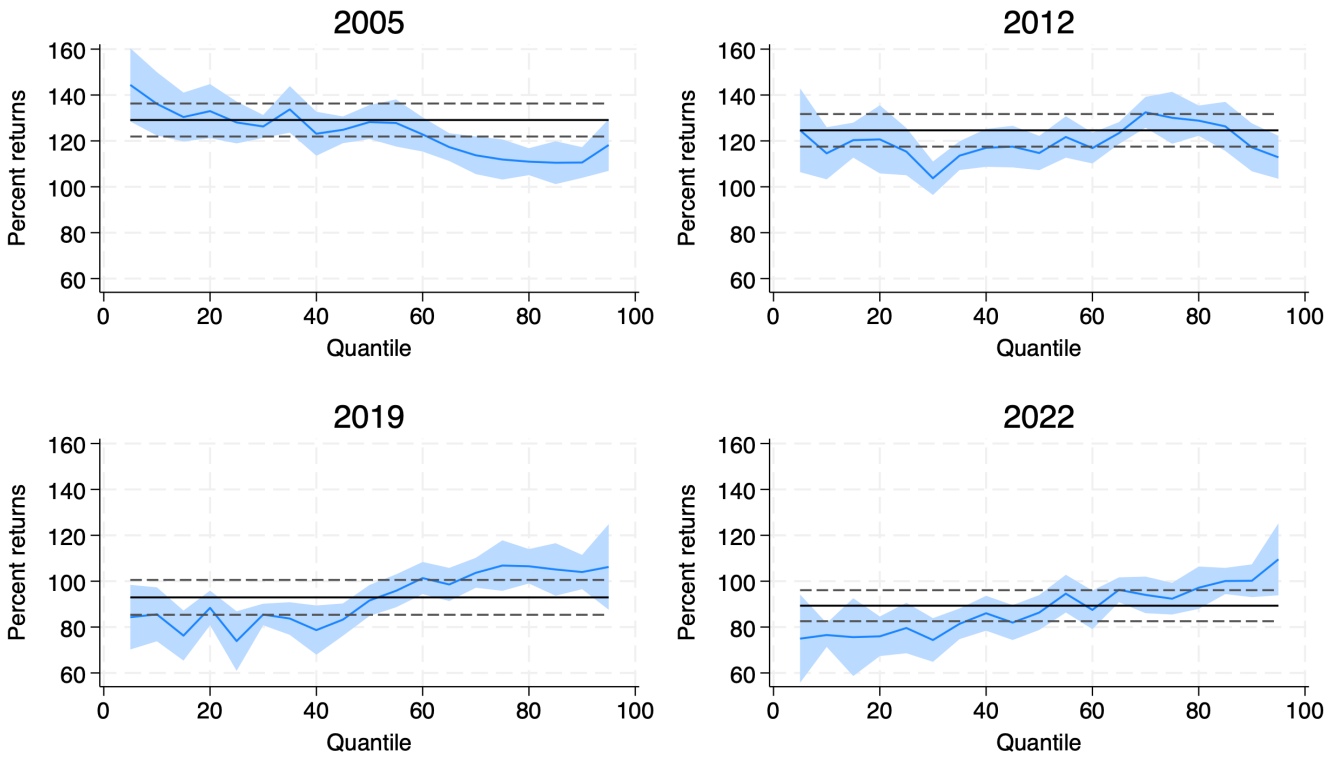
Note: Regression coefficients are multiplied by 100 percent. The lines represent 95 percent confidence intervals.
Source: Authors' computations from the 2005, 2012, 2019, and 2020–2022 October rounds of the Labor Force Survey (PSA various years)

Table 3. Returns to education: Conditional quantile regression, full sample, 2005 and 2022

	2005			2022			Change for Q10 (pp)	Change for Q90 (pp)
	Q10	Q50	Q90	Q10	Q50	Q90		
Elementary	0.237***	0.269***	0.195***	0.175***	0.211***	0.149***	-6.2	-4.6
	(0.025)	(0.017)	(0.019)	(0.031)	(0.018)	(0.010)		
Secondary	0.479***	0.569***	0.433***	0.343***	0.361***	0.345***	-13.6	-8.8
	(0.027)	(0.017)	(0.019)	(0.040)	(0.021)	(0.027)		
Partial college	0.764***	0.837***	0.680***	0.440***	0.418***	0.541***	-32.4	-13.9
	(0.032)	(0.017)	(0.021)	(0.036)	(0.022)	(0.023)		
College	1.440***	1.410***	1.253***	0.800***	0.939***	1.071***	-64.0	-18.2
	(0.027)	(0.016)	(0.021)	(0.034)	(0.021)	(0.015)		
N	30,891	30,891	30,891	37,179	37,179	37,179		

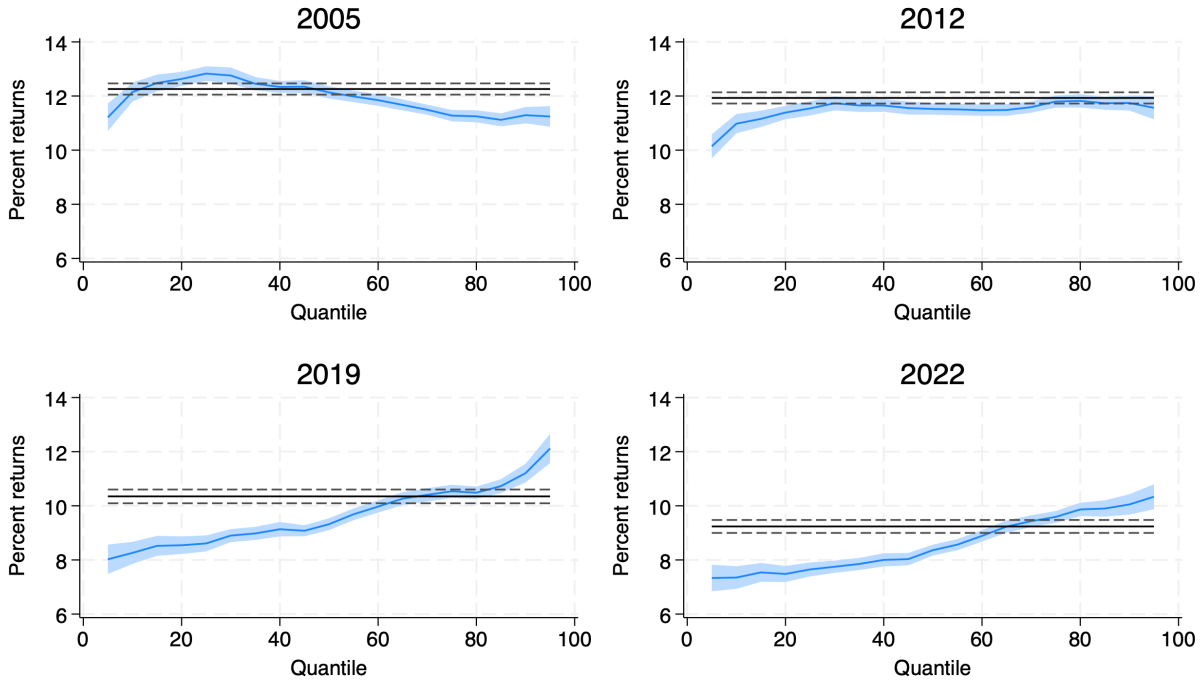
Notes: *** $p<.01$, ** $p<.05$, * $p<.1$. pp = percentage point; Q = quantile. Other controls not shown: sex, experience, squared experience.
Source: Authors' computations from the 2005, 2012, 2019, and 2020–2022 October rounds of the Labor Force Survey (PSA various years)

Figure 5. Returns to college education by quantile, young sample, 2005–2022



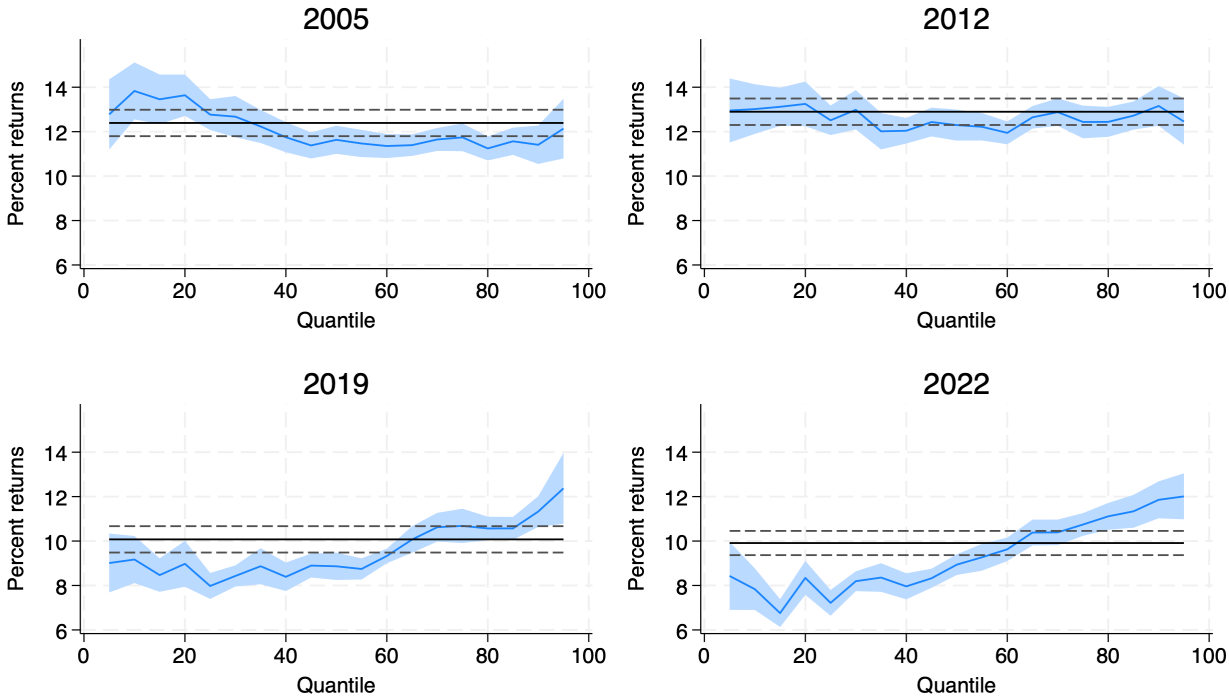
Note: Regression coefficients are multiplied by 100 percent. The lines represent 95 percent confidence intervals.
Source: Authors' computations from the 2005, 2012, 2019, and 2022 October rounds of the Labor Force Survey (PSA various years)

Figure 6. Returns to a year of schooling by quantile, full sample, 2005–2022



Note: Regression coefficients are multiplied by 100 percent. Data show percent increase in hourly real wages based on an extra year of schooling. The lines represent 95 percent confidence intervals.
Source: Authors' computations from the 2005, 2012, 2019, and 2022 October rounds of the Labor Force Survey (PSA various years)

Figure 7. Returns to a year of schooling by quantile, young sample, 2005–2022



Note: Regression coefficients are multiplied by 100 percent. Data show percent increase in hourly real wages based on an extra year of schooling. The lines represent 95 percent confidence intervals.
Source: Authors' computations from the 2005, 2012, 2019, and 2022 October rounds of the Labor Force Survey (PSA various years)

Factors contributing to declining returns

The decline in returns to education in the Philippines may be attributed to several factors that have shaped the labor market in recent years.

Increased formal employment and a shift from agriculture to services

Figure 8 illustrates highest grade completed by sector of employment for the young sample in 2005, 2019, and 2022. The proportion of college graduates increased across sectors, particularly in services. Formal employment also increased by 13.91 percentage points between 2005 and 2019. Furthermore, there is a significant shift from agriculture to services.

Another notable factor to consider is the increased level of formality in employment. Figure 9 illustrates the types of employment within the young sample, showing a substantial rise in formal employment between 2005 and 2019. Specifically, the proportion of workers in formal jobs increased by 13.91 percentage points during this period.

Figure 10 depicts the employment by sector of salaried workers in the young sample. It reveals a significant sectoral shift, particularly the movement from agriculture to services. Among salaried workers, there was an increase of 4.79 percentage points in services jobs, almost equal to the decline of 4.69 percentage points in agricultural employment. This shift has implications for the variations in the returns to education across different sectors.

Quality of jobs created

Moreover, the quality of jobs created during this period seems to have contributed to the declining returns. Figure 11 shows that median wages for individuals with lower education levels improved, narrowing the wage gap between them and college graduates.

Quality of education

The rapid expansion of access to education may have come at the cost of maintaining educational quality. As enrollment rates surged, particularly in higher education, the education system faced challenges, such as inadequate resources, overcrowded classrooms, and a lack of qualified teachers (ADB 2020). These factors can lead to suboptimal learning outcomes, ultimately reducing the value of education in the labor market.

Moreover, concerns have arisen regarding the relevance of educational programs to the evolving needs of industries. The misalignment between the skills and knowledge imparted by educational institutions and labor market demands can contribute to the declining returns to education (ADB 2020).

How can educational returns be enhanced for economic equity?

Recognize the tangible benefits of improving educational access

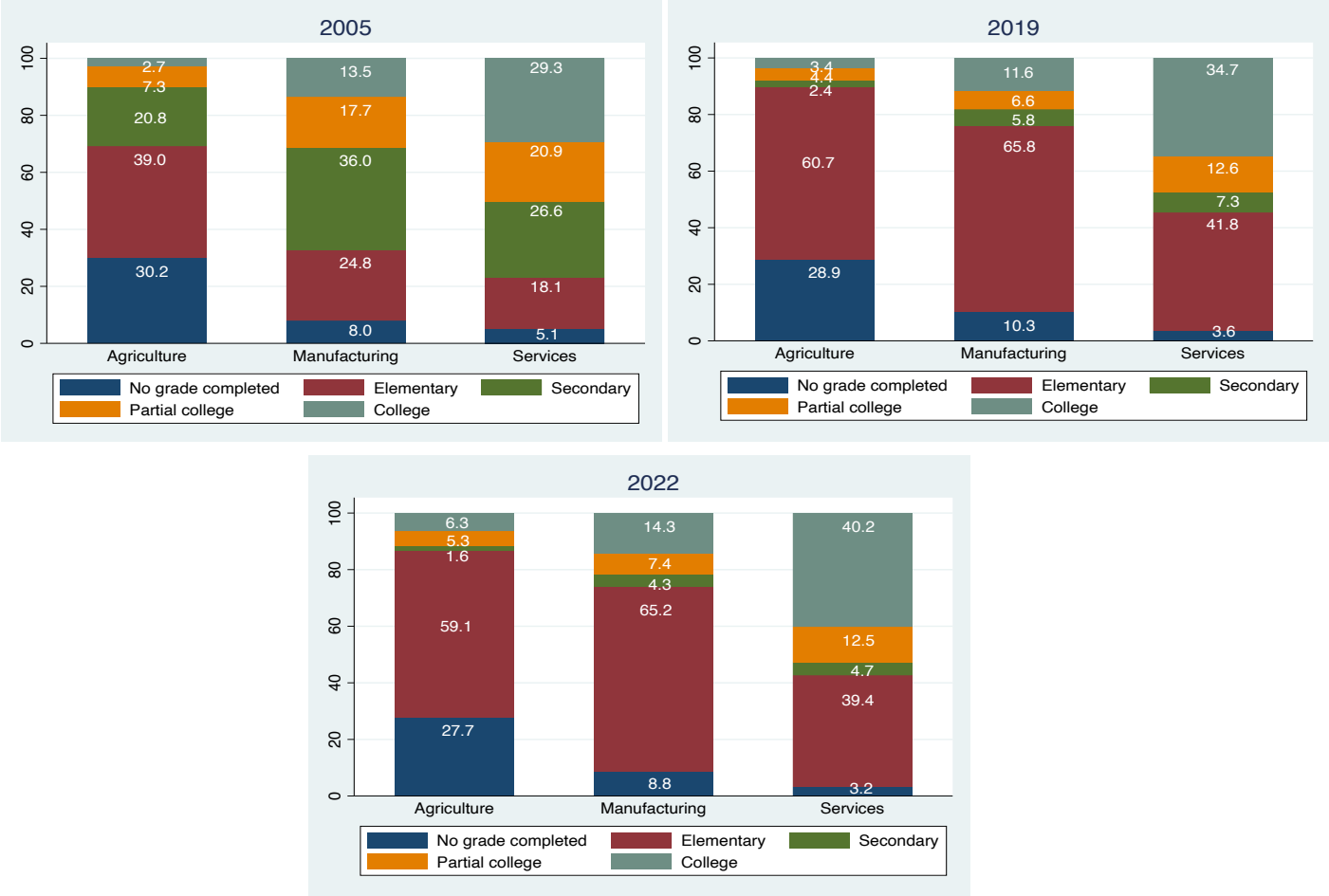
Although there may be some convergence of returns across educational levels, it is important to note that higher levels of schooling, especially college education, still offer superior outcomes in terms of employment prospects and earnings potential. Therefore, in addition to efforts to expand access to education, policies must also focus on enabling individuals to pursue higher levels of education.

Address inequities among college graduates

The widening inequities in returns among graduates, especially college graduates, require targeted policy interventions. Several key focus areas emerge from the analysis:

a. ***Ensure quality in higher education.*** The Commission on Higher Education (CHED) plays a pivotal role in strengthening its capacity to monitor and enforce quality standards across the country's 2,396 higher educational institutions (HEIs). This will help ensure that graduates possess the skills and competencies needed to boost their productivity and secure better employment opportunities.

Figure 8. Highest grade completed by sector of employment (young sample): 2005, 2019, 2022



Source: Authors' computations from the 2005, 2012, 2019, and 2022 October rounds of the Labor Force Survey (PSA various years)

Figure 9. Highest grade completed among individuals employed in the formal and informal sectors, unemployed, and not in the labor force by survey year (young sample): 2005, 2019, 2022

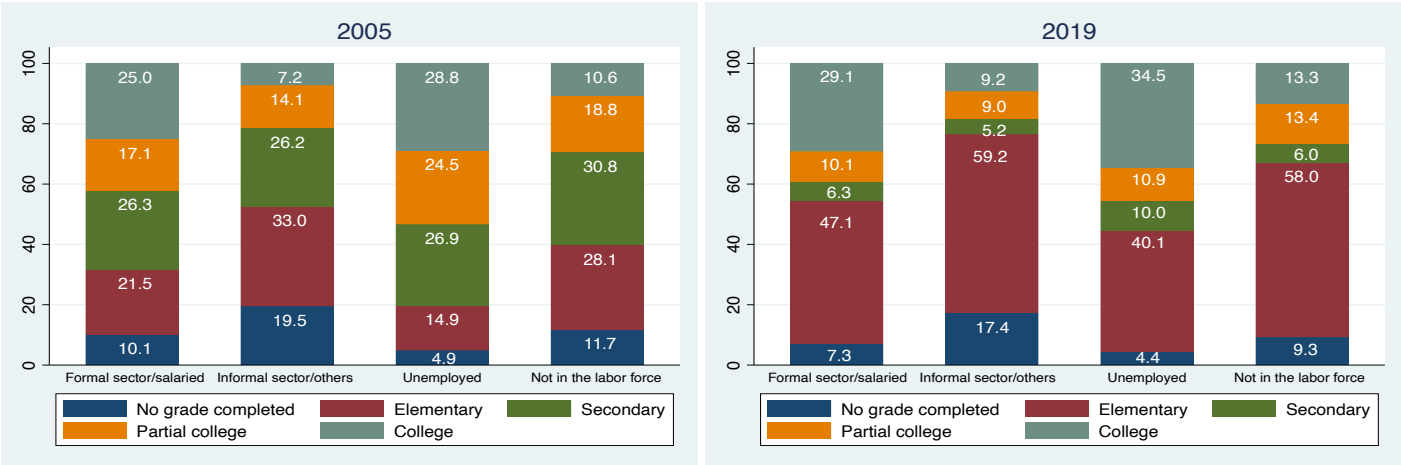
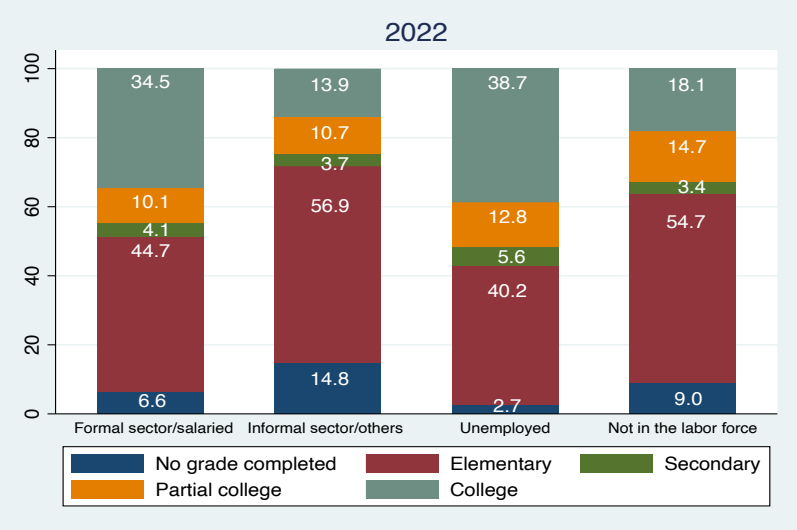
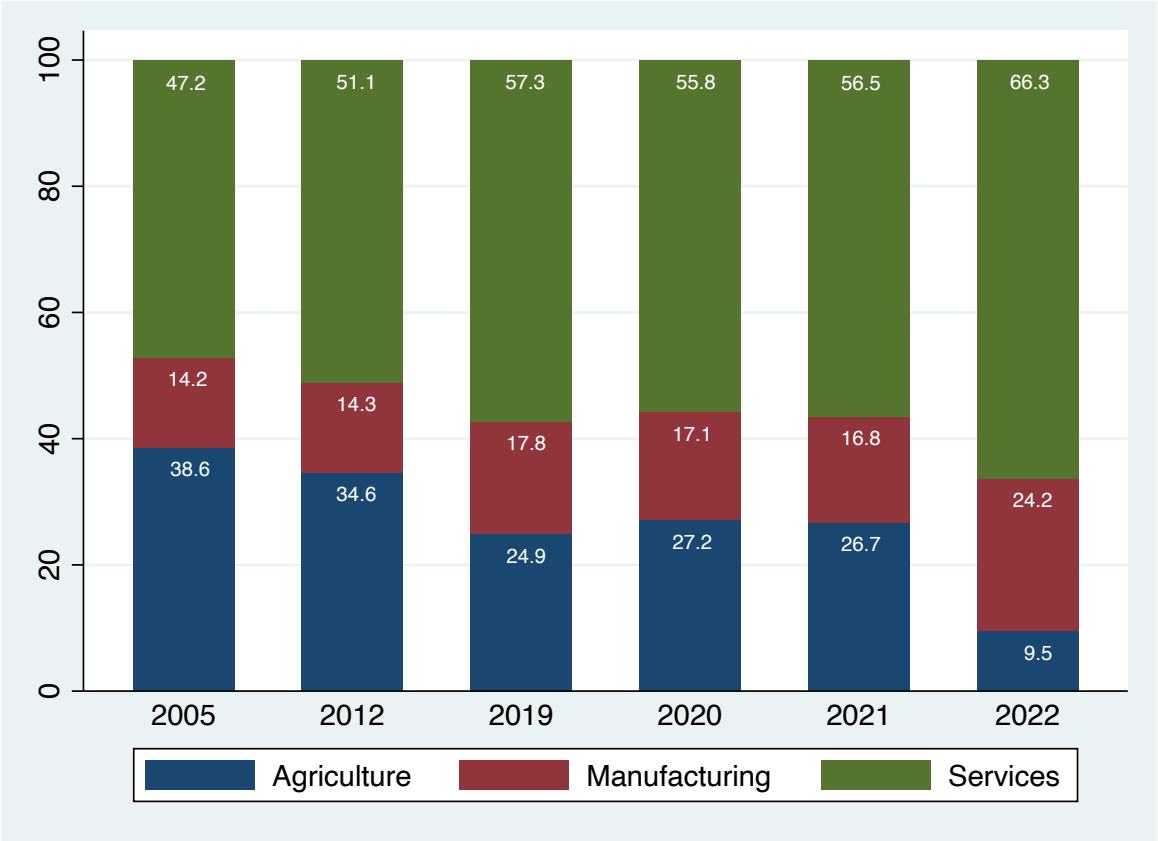


Figure 9 *(continued)*



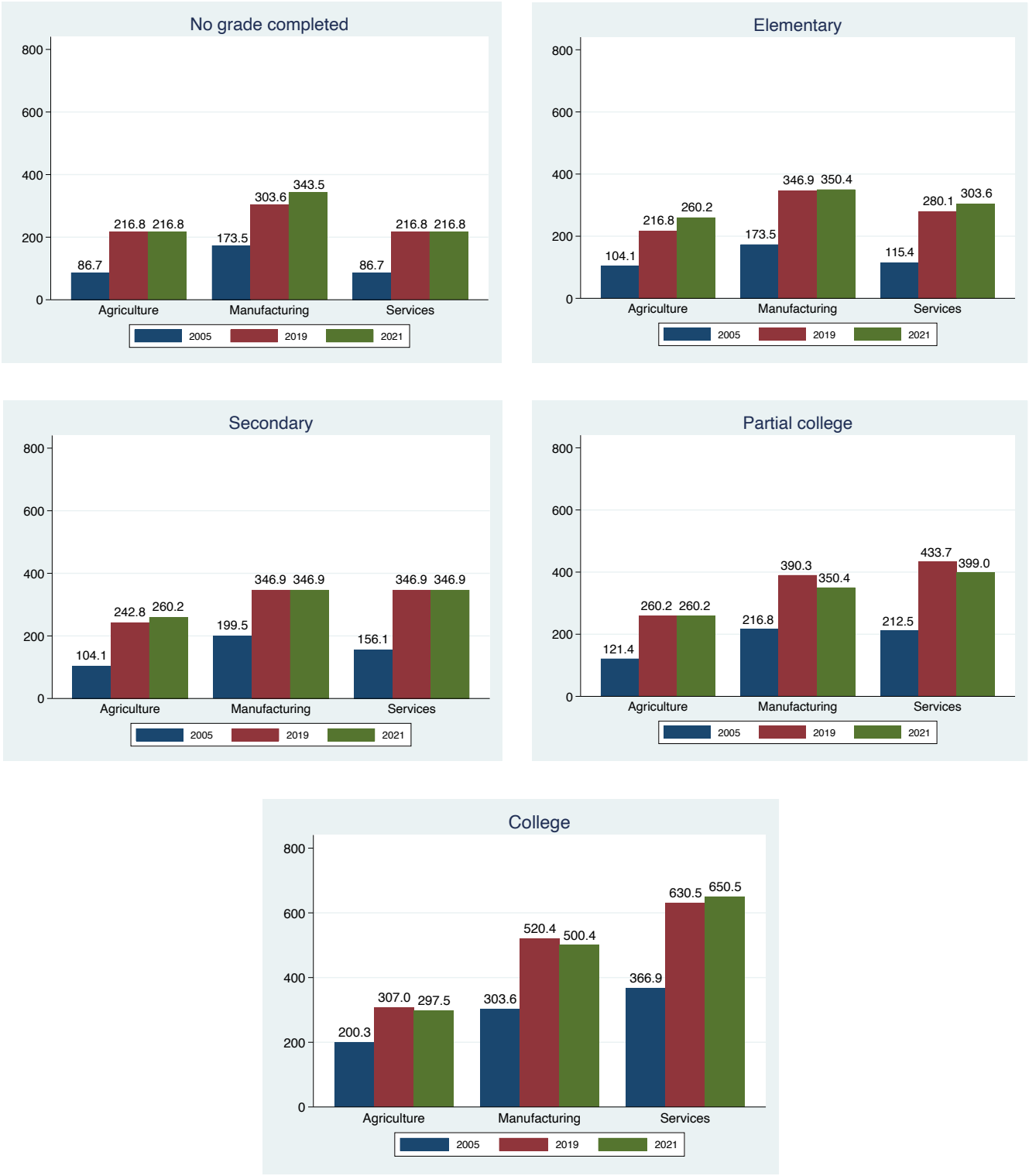
Source: Authors' computations from the 2005, 2012, 2019, and 2022 October rounds of the Labor Force Survey (PSA various years)

Figure 10. Employment of salaried workers by sector (young sample)



Source: Authors' computations from the 2005, 2012, 2019, and 2020–2022 October rounds of the Labor Force Survey (PSA various years)

Figure 11. Median daily wages by highest educational attainment by sector of employment: 2005, 2019, 2021



Source: Authors' computations from the 2005, 2012, 2019, and 2020–2022 October rounds of the Labor Force Survey (PSA various years)

b. ***Provide information on institutional quality.*** To support informed decisionmaking, CHED should establish an accessible database that provides comprehensive information on the performance of HEIs, including metrics such as graduate employment rates, average alumni salaries, and completion rates.

c. ***Implement financial support mechanisms.*** It is essential to fully implement the Student Loan Program for Tertiary Education to ensure equitable access to higher education. This program will enable students to attend their college of choice regardless of their socioeconomic background. By removing financial barriers, this program can help level the playing field and promote greater equity in higher education opportunities.

Align higher education with industry needs

Aligning higher education with industry needs is crucial for maximizing economic returns. HEIs should prioritize developing diverse academic programs that cater to evolving industry requirements, particularly in manufacturing and services. This alignment becomes even more critical in the context of increased technological adoption and the transition toward a greener economy. Close collaboration with industry partners can ensure that graduates possess the relevant skills and knowledge demanded by the labor market.

Align educational expansion with economic measures

Educational expansion alone is insufficient to guarantee improved outcomes. Parallel economic measures are necessary to absorb the growing number of educated individuals into quality jobs. Policymakers must prioritize creating a climate that attracts investment and stimulates growth in sectors requiring highly skilled workers. This involves implementing policies and incentives that encourage the development of high-skill jobs, fostering innovation, and promoting entrepreneurship. Removing regulatory barriers and creating a conducive business environment for knowledge-intensive industries is equally important. Without such efforts, the risk of brain drain increases as college graduates seek employment overseas, potentially further depressing domestic returns to education.

Enhance education quality

Alongside expanding access to education, policymakers must prioritize improving the quality of education at all levels. This can be achieved through the following mechanisms:

- Adequate funding and resource allocation, ensuring well-equipped classrooms, qualified teachers, and appropriate learning materials, particularly in underserved areas;
- Regular curriculum review and updating, which are necessary to align with industry needs and global standards to better prepare graduates for successful careers;
- Investing in continuous professional development for educators, which is crucial for enhancing teaching practices and adapting to evolving educational technologies; and
- Strengthening quality assurance mechanisms and accreditation systems that will help maintain consistent, high-quality education across institutions, with regular monitoring and evaluation identifying areas for improvement and maintaining academic standards.

Conclusion and future research directions

The authors find empirical evidence of declining but converging returns to education in the Philippines from 2005 to 2022. Returns to higher education declined the most, but especially for low-wage workers. This suggests widening economic inequities among college graduates.

Further research avenues should be pursued to deepen understanding of these dynamics. Future studies should aim to analyze the causal factors that may explain the declining returns to education. They should focus on investigating the variances in returns to college education, taking into account factors such as the quality and prestige of HEI attended, the field of study pursued, and the type of employment (public or private).



"Policymakers must prioritize creating a climate that attracts investment and stimulates growth in sectors requiring highly skilled workers. This involves implementing policies and incentives that encourage the development of high-skill jobs, fostering innovation, and promoting entrepreneurship."

Understanding how these factors interact with returns to education can provide valuable insights for policymakers and inform targeted interventions to address inequities. This could involve longitudinal studies tracking graduates from different institutions and fields over time to assess long-term career outcomes and earnings trajectories.

As the Philippine education system undergoes significant transformations, it is crucial to assess the impact of recent policy changes on returns to education and inequality. Future research should explore the effects of introducing senior high school in 2016 and expanding state and locally funded HEIs from 2018 onwards. This could include comparative analyses of cohorts before and after these policy implementations, as well as qualitative studies to understand stakeholders' perspectives on these changes. Additionally, investigating the impact of the COVID-19 pandemic on education quality and subsequent labor market outcomes would provide timely insights for policy adaptation amid global challenges.

Further research should also extend data collection and analysis efforts to include informal workers and overseas Filipino workers comprehensively, as they constitute a significant portion of the labor force. By capturing the experiences and outcomes of these groups, researchers can paint a more complete picture of the returns to education and the factors influencing labor market dynamics in the Philippines.

Policymakers must prioritize education quality, relevance, and accessibility while creating high-skill jobs aligned with the growing educated workforce. As the Philippines navigates post-pandemic challenges, leveraging human capital potential is crucial for inclusive, sustainable growth. Future research should allow policymakers and educators to gain a more nuanced understanding of the education-labor market nexus in the Philippines, enabling more effective and targeted interventions to enhance educational outcomes and economic opportunities for all.

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